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Dear Sir/Madam,

Please consider this recommendation of Dr. Akul Yugesh Mehta and his outstanding work in the field of biotechnologies. Dr. Mehta received his B.Pharm. in pharmaceutical sciences from the University of Mumbai in 2007, which he followed with his Ph.D. in the same field from Virginia Commonwealth University in 2014. He completed 2 years of postdoctoral study there, before accepting a position as a research scholar at the Beth Israel Deaconess Medical Center affiliated with Harvard Medical School, both in their Department of Surgery and the National Center for Functional Glycomics. In 2021, Dr. Mehta accepted his current position of instructor of surgery, staff scientist I there. Through this position, Dr. Mehta has been able to continue his outstanding work centered on the development of improved approaches for studying glycans and their interactions with proteins.

As for myself, I am the vice president of product at Vector Laboratories, Inc. I have worked here for 15 years, and my current position follows my prior positions of senior scientist, director of R&D, and senior director of both R&D and operations. I received my B.S. in biochemistry from the California Polytechnic State University – San Luis Obispo and my Ph.D. in immunology from the Chan Medical School at the University of Massachusetts. As a part of research at Vector Laboratories, Inc., we make use of various tools produced by Dr. Mehta for studying glycans. Due to this, I have become familiar with both Dr. Mehta and his research.

Glycans are carbohydrate biological molecules with extensive value for many key goals in efforts to improve human health and disease treatment, including in treating HIV, COVID-19, cancer, and auto-immune diseases. Because of the major costs associated with producing pure glycans, greater than $5000 for 1 mg in some cases, and the necessity of screening and testing these glycans hundreds of times, Dr. Mehta has developed a series of highly miniaturized microarrays that have enabled researchers to work with these materials despite the high costs. Specifically, he has produced more than 7000 microarrays divided into 20 types in which picograms of glycans are printed onto glass slides in microscopic spots. These arrays are then tested against biological agents, which enables the collection of highly valuable data while using a minimal amount of glycans. Dr. Mehta has tailored and produced these microarrays and applied them in a number of areas, including studies of natural human glycans centered on developing an improved understanding of allergies and studies centered on the development of HIV neutralizing antibodies. His developments are relevant to checking the quality of certain reagents and gaining information about new reagents produced by our company. Throughout these studies, Dr. Mehta has produced highly efficient tools that have proven invaluable for supporting these critical research efforts.

Beyond his microarrays, Dr. Mehta has also produced impressive software tools with excellent applications for studying glycans. One tool, which Dr. Mehta named GLAD (developed using HTML, CSS, and JavaScript), provides the capacity to effectively visualize and perform data analysis on glycan microarrays. He designed this tool to present a variety of visualizations, including heat maps, force graphs, and bar charts and

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it works on any web browser, thereby making it highly accessible to all scientists. He also developed another web-based tool known as GlycoGlyph capable of effectively representing complex glycan structures and converting between various forms of glycan nomenclature. These tools have proven highly useful for many researchers working with glycans and have facilitated a substantial number of breakthroughs. Dr. Mehta is also a prominent contributor to bioinformatics in glycosciences and has even organized seminar series with the Glycoinformatics Consortium (GLIC) with a special topic for glycan arrays. He is also a contributor of MIRAGE Commission, which helps develop data reporting guidelines for glycomics experiments, particularly glycan array experiments.

Glycans are among the most important biological molecules currently under investigation, as noted above, as they play an important role in a variety of diseases. For example, the COVID-19 virus makes use of spike proteins coated in glycans in order to infect humans and evade immune responses. Some forms of cancers also use glycan coats to similarly shield their cells from the immune system. Even the ABO blood groups are glycans, and play an important role in transfusion medicine. By investigating these glycans, researchers are steadily unlocking improved approaches for breaking through these barriers. Throughout his career, Dr. Mehta has demonstrated himself to be uniquely talented in producing improved tools for studying these molecules. His work has proven crucial for a multitude of researchers from around the world, whose findings have had profound implications for improving the treatment of these conditions. The benefits of Dr. Mehta's continued research efforts are therefore quite clear.

Before concluding, I would also like to briefly mention Dr. Mehta's extensive history of peer review service for a variety of journals, including *Cancers, Bioinformatics,* and *Chemical Communications*. Due to the essential nature of this service, as peer review is the primary method for establishing the validity and originality of a piece of research, these positions are typically reserved for well-established experts on subject matter relevant to the journal. Dr. Mehta's service in this regard is therefore a clear acknowledgement of his expertise on the part of these journals.

Based on everything discussed throughout this letter, the importance of Dr. Mehta and his research for the field of biotechnologies is well-established. I therefore offer you my highest recommendation of Dr. Mehta and urge you to support his efforts to continue his research without interruption.

Sincerely,

Pamela James

Dr. Pamela James

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